Topologies & Network Devices(Hubs, Switches, Router, Gateway)

Lec-2 Networks

What is Topology

- The arrangement of a network which comprises of nodes and connecting lines via sender and receiver is referred as network topology
- Mesh
- Star
- Bus
- Ring
- Hybrid

Mesh

- Every device is connected with another via dedicated channels. These channels are known as links.
- If suppose, N number of devices are connected with each other in mesh topology, then total number of ports that is required by each device is ? N-1. In the Figure 1, there are 5 devices connected to each other, hence total number of ports required is 4.
- If suppose, N number of devices are connected with each other in mesh topology, then total number of dedicated links required to connect them is ${}^{N}C_{2}$ i.e. N(N-1)/2. In the Figure 1, there are 5 devices connected to each other, hence total number of links required is 5*4/2 = 10.

Star

- In star topology, all the devices are connected to a single hub through a cable. This hub is the central node and all others nodes are connected to the central node. The hub can be passive ?in nature i.e. not intelligent hub such as broadcasting devices, at the same time the hub can be intelligent known as active ?hubs. Active hubs have repeaters in them. If N devices are connected to each other in star topology, then the number of cables required to connect them is N. So, it is easy to set up.
- Each device require only 1 port i.e. to connect to the hub.
- If the concentrator (hub) on which the whole topology relies fails, the whole system will crash down.

Bus

- Bus topology is a network type in which every computer and network device is connected to single cable. It transmits the data from one end to another in single direction. No bi-directional feature is in bus topology.
- If N devices are connected to each other in bus topology, then the number of cables required to connect them is 1 ?which is known as backbone cable and N drop lines are required.
- Cost of the cable is less as compared to other topology, but it is used to built small networks.
- If the common cable fails, then the whole system will crash down.
- If the network traffic is heavy, it increases collisions in the network. To avoid this, various protocols are used in MAC layer known as Pure Aloha, Slotted Aloha, CSMA/CD etc.

Ring

- In this topology, it forms a ring connecting a devices with its exactly two neighboring devices. One station is known as **monitor** station which takes all the responsibility to perform the operations.
- To transmit the data, station has to hold the token. After the transmission is done, the token is to be released for other stations to use.
- When no station is transmitting the data, then the token will circulate in the ring.
- There are two types of token release techniques : **Early token release** releases the token just after the transmitting the data and **Delay token release** releases the token after the acknowledgement is received from the receiver.
- The possibility of collision is minimum in this type of topology.

Hybrid

- This topology is a collection of two or more topologies

Network Devices (Hub, Repeater, Bridge, Switch, Router, Gateways)

- **1. Repeater** – A repeater operates at the physical layer. Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted so as to extend the length to which the signal can be transmitted over the same network. An important point to be noted about repeaters is that they do not amplify the signal. When the signal becomes weak, they copy the signal bit by bit and regenerate it at the original strength. It is a 2 port device.

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- **Hub** – A hub is basically a multiport repeater. A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations. Hubs cannot filter data, so data packets are sent to all connected devices. In other words, collision domain of all hosts connected through Hub remains one. Also, they do not have intelligence to find out best path for data packets which leads to inefficiencies and wastage.
- **Types of Hub**
- **Active Hub:-** These are the hubs which have their own power supply and can clean, boost and relay the signal along with the network. It serves both as a repeater as well as wiring center. These are used to extend the maximum distance between nodes.
- **Passive Hub :-** These are the hubs which collect wiring from nodes and power supply from active hub. These hubs relay signals onto the network without cleaning and boosting them and can’t be used to extend the distance between nodes.

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- **Bridge** – A bridge operates at data link layer. A bridge is a repeater, with add on the functionality of filtering content by reading the MAC addresses of source and destination. It is also used for interconnecting two LANs. It has a single input and single output port, thus making it a 2 port device.
- **Types of Bridges**
- **Transparent Bridges:-** These are the bridge in which the stations are completely unaware of the
  bridge's existence i.e. whether or not a bridge is added or deleted from the network, reconfiguration of
  the stations is unnecessary. These bridges make use of two processes i.e. bridge forwarding and bridge learning.
- **Source Routing Bridges:-** In these bridges, routing operation is performed by source station and the frame specifies which route to follow. The hot can discover frame by sending a special frame called discovery frame, which spreads through the entire network using all possible paths to destination.

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- **Switch** – A switch is a multiport bridge with a buffer and a design that can boost its efficiency(a large number of ports imply less traffic) and performance. A switch is a data link layer device. The switch can perform error checking before forwarding data, that makes it very efficient as it does not forward packets that have errors and forward good packets selectively to correct port only. In other words, switch divides collision domain of hosts, but broadcast domain remains same.

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- **Routers** – A router is a device like a switch that routes data packets based on their IP addresses. Router is mainly a Network Layer device. Routers normally connect LANs and WANs together and have a dynamically updating routing table based on which they make decisions on routing the data packets. Router divide broadcast domains of hosts connected through it.

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- **Gateway** – A gateway, as the name suggests, is a passage to connect two networks together that may work upon different networking models. They basically work as the messenger agents that take data from one system, interpret it, and transfer it to another system. Gateways are also called protocol converters and can operate at any network layer. Gateways are generally more complex than switch or router.

Cables,

- Different types of network cables, such as coaxial cable, optical fiber cable, and twisted pair cables, are used depending on the network's physical layer, topology, and size
- The devices can be separated by a few meters (e.g. via Ethernet) or nearly unlimited distances (e.g. via the interconnections of the Internet).

Twisted pair

- Twisted pair cabling is a form of wiring in which pairs of wires (the forward and return conductors of a single circuit) are twisted together for the purposes of canceling out electromagnetic interference (EMI) from other wire pairs and from external sources. This type of cable is used for home and corporate Ethernet networks. Twisted pair cabling is used in short patch cables and in the longer runs in structured cabling.
- An Ethernet crossover cable is a type of twisted pair Ethernet cable used to connect computing devices together directly that would normally be connected via a network switch, Ethernet hub or router, such as directly connecting two personal computers via their network adapters. Most current Ethernet devices support Auto MDI-X, so it doesn't matter whether you use crossover or straight cables

Fiber optics

- An optical fiber cable consists of a center glass core surrounded by several layers of protective material. Optical fiber deployment is more expensive than copper but offers higher bandwidth and can cover longer distances.[2] There are two major types of optical fiber cables: shorter-range multi-mode fiber and long-range single-mode fiber.

Coaxial

- Coaxial cables form a transmission line and confine the electromagnetic wave inside the cable between the center conductor and the shield. The transmission of energy in the line occurs totally through the dielectric inside the cable between the conductors. Coaxial lines can therefore be bent and twisted (subject to limits) without negative effects, and they can be strapped to conductive supports without inducing unwanted currents in them.
- Early Ethernet, 10BASE5 and 10BASE2, used baseband signaling over coaxial cables. In the 20th century the L-carrier system used coaxial cable for long-distance calling.
- Coaxial cables are commonly used for television and other broadband signals. Although in most homes coaxial cables have been installed for transmission of TV signals, new technologies (such as the ITU-T G.hn standard) open the possibility of using home coaxial cable for high-speed home networking applications (Ethernet over coax).

Patch

- A patch cable is an electrical or optical cable used to connect one electronic device to another towards building infrastructure for signal routing. Devices of different types (e.g. a switch connected to a computer, or a switch connected to a router) are connected with patch cables. Patch cables are usually produced in many different colors so as to be easily distinguishable,[1] and most are relatively short, no longer than a few meters. In contrast to on-premises wiring, patch cables are more flexible.